

HEATSHIELD AI

Extreme Heatwave Early Warning & Human Thermal Stress Decision Intelligence Platform

Sponsoring Ministry:
Ministry of Earth Sciences (MoES)

Department & Target:
NCMRWF / ULBs, DDMAAs, PHCs

National Alignment:
Mission Mausam & NDMA Guidelines

1. The Winning Positioning & Executive Pitch

30-Second Elevator Pitch

"Traditional heat alerts report shade temperature for an entire district, ignoring the lethal compounding effects of humidity, solar radiation, and concrete heat trapping. When humidity is high, human evaporative cooling collapses, causing fatal heat stroke even at 38°C.

*HeatShield AI shifts national forecasting from 'what the weather will be' to 'what the weather will do to human physiology at the municipal ward level.' By coupling NCMRWF numerical forecasts with satellite Land Surface Temperature and vegetation canopy deficit, we compute true physiological stress (**UTCI** for pedestrians and **WBGT** for outdoor laborers), explain the risk drivers using **TreeSHAP**, and close the loop with **automated municipal logistics dispatch**—directly pushing water tanker routes, cooling shelter orders, and labor curfews to officers' phones 3 to 5 days ahead."*

The 6 Critical Differentiators

Differentiator	Technical Execution & Strategic Value
1. Native NCMRWF Integration	Directly ingests NCMRWF NCUM 12km model grids and IMD AWS GRIB2/NetCDF formats rather than relying on generic commercial APIs.
2. Analytical $\$T_{\text{mrt}}\$$ Derivation	Solves Mean Radiant Temperature via ISO 7726 solar-geometric equations and Sentinel-2 albedo, bypassing the physical lack of black-globe thermometers.
3. Dual Physiological Standards	Computes UTCI for the general public/pedestrians and ISO 7243 WBGT for occupational outdoor workers (police, construction, sanitation).
4. Transparent Explainability	Uses TreeSHAP to inform administrators <i>why</i> a ward is red (e.g., +28% UTCI, +18% tin-roof LST anomaly, +12% canopy deficit).
5. Closed-Loop Dispatch	Automated webhook pushes a structured NDMA Municipal Action Ticket live to an evaluator's WhatsApp during live pitch.
6. Epidemiological Grounding	Calibrates risk against peer-reviewed Distributed Lag Non-linear Models (DLNM) from Ahmedabad and Delhi heatwave epidemiological studies.

2. The Scientific Core & Mathematical Formulations

2.1 The Biophysical Energy Balance

Human thermal survival is governed by the thermodynamic heat exchange between the human core and the atmospheric boundary layer:

$S = M - W - (R + C + E + RES)$

- **S**: Rate of body heat storage (If $S > 0$, core body temperature rises rapidly).
- **M**: Metabolic heat production (~135 W/m² walking; up to 300 W/m² for construction labor).
- **W**: Mechanical work performed by the body.
- **R**: Net radiant heat flux (from direct sun, sky, asphalt, and concrete).
- **C**: Convective heat flux (air temperature and surface boundary layer wind speed).
- **E**: Evaporative heat loss from sweat:

$E = (p_{sk,s} - p_a) / (R_{e,cl} + 1/h_e)$

When relative humidity breaches 75–80% in high heat, ambient vapor pressure p_a approaches skin saturation pressure $p_{sk,s}$. Evaporative cooling collapses ($E \rightarrow 0$), spiking core temperature past 40.5°C and triggering heat stroke.

- **RES**: Heat loss via respiration (latent and sensible).

2.2 Analytical Mean Radiant Temperature (ISO 7726)

Without black-globe sensors across city streets, HeatShield AI derives T_{mrt} analytically via solar-geometric energy balance:

$T_{mrt} = [(1/\sigma) \cdot (F_{sky} \epsilon_{sky} \sigma T_a^4 + F_{gnd} \epsilon_{gnd} \sigma T_{sfc}^4 + (f_p \cdot I_{dir} / \cos(\theta)) + F_{sky} \cdot I_{diff})]^{0.25} - 273.15$

- $\sigma = 5.67 \times 10^{-8}$ W/m²K⁴ (Stefan-Boltzmann constant).
- T_a : Ambient air temperature (K); T_{sfc} : Satellite Land Surface Temperature from INSAT-3DR/Landsat (K).
- I_{dir}, I_{diff} : Direct normal and diffuse horizontal solar irradiance (W/m²).
- θ : Solar zenith angle calculated from latitude, longitude, day of year, and UTC hour.
- f_p : Projected human area factor: $f_p = 0.308 \cdot \cos(\theta) \cdot (1 - \sin(\theta)) + 0.053$
- F_{sky}, F_{gnd} : Sky and ground view factors (~0.5 in flat urban terrain).

2.3 Vectorized UTCI Polynomial Engine

Operationalized via the WMO-approved 6th-order Taylor polynomial regression approximating the Fanger multi-node model:

$UTCI = T_a + \Delta UTCI(T_a, v_{10}, e, T_{mrt} - T_a)$

Where vapor pressure e (hPa) is computed via the Magnus-Tetens formula:

$e = (RH / 100) \cdot 6.112 \cdot \exp(17.67 \cdot T_a / (T_a + 243.5))$

UTCI Range (°C)	Stress Category	Physiological & Clinical Impact
< 9 to 26	No Thermal Stress	Normal thermal comfort; homeostatic equilibrium.
26 to 32	Moderate Heat	Slight increase in sweat rate; mild cardiovascular load.
32 to 38	Strong Heat	Heavy sweating; vasodilation; cardiovascular fatigue in elderly.

UTCI Range (°C)	Stress Category	Physiological & Clinical Impact
38 to 46	Very Strong Heat	Evaporative deficit; severe dehydration hazard; clinical heat exhaustion.
> 46	Extreme Heat	Acute hyperthermia risk; core temp > 40.5°C; cellular damage & heat stroke.

2.4 Occupational WBGT (ISO 7243)

For outdoor laborers (traffic police, construction, sanitation), WBGT is computed under ISO 7243:

$$WBGT_{outdoor} = 0.7 \cdot T_{nw} + 0.2 \cdot T_g + 0.1 \cdot T_a$$

Where globe temperature $T_g \approx 0.01498 \cdot T_{mrt} + 0.7302 \cdot T_a + 0.009 \cdot RH - 1.38 \cdot \sqrt{v}$

Statutory Work-to-Rest Cycle Guidelines (ISO 7243):

- **WBGT < 28°C:** Continuous normal work (100%).
- **WBGT 28°C – 30°C:** 75% work, 25% rest per hour in shade; mandatory hydration breaks.
- **WBGT 30°C – 32°C:** 50% work, 50% rest per hour; emergency water stations deployed.
- **WBGT ≥ 32°C: Statutory work stoppage (Curfew)** for outdoor physical labor (11:30–15:30).

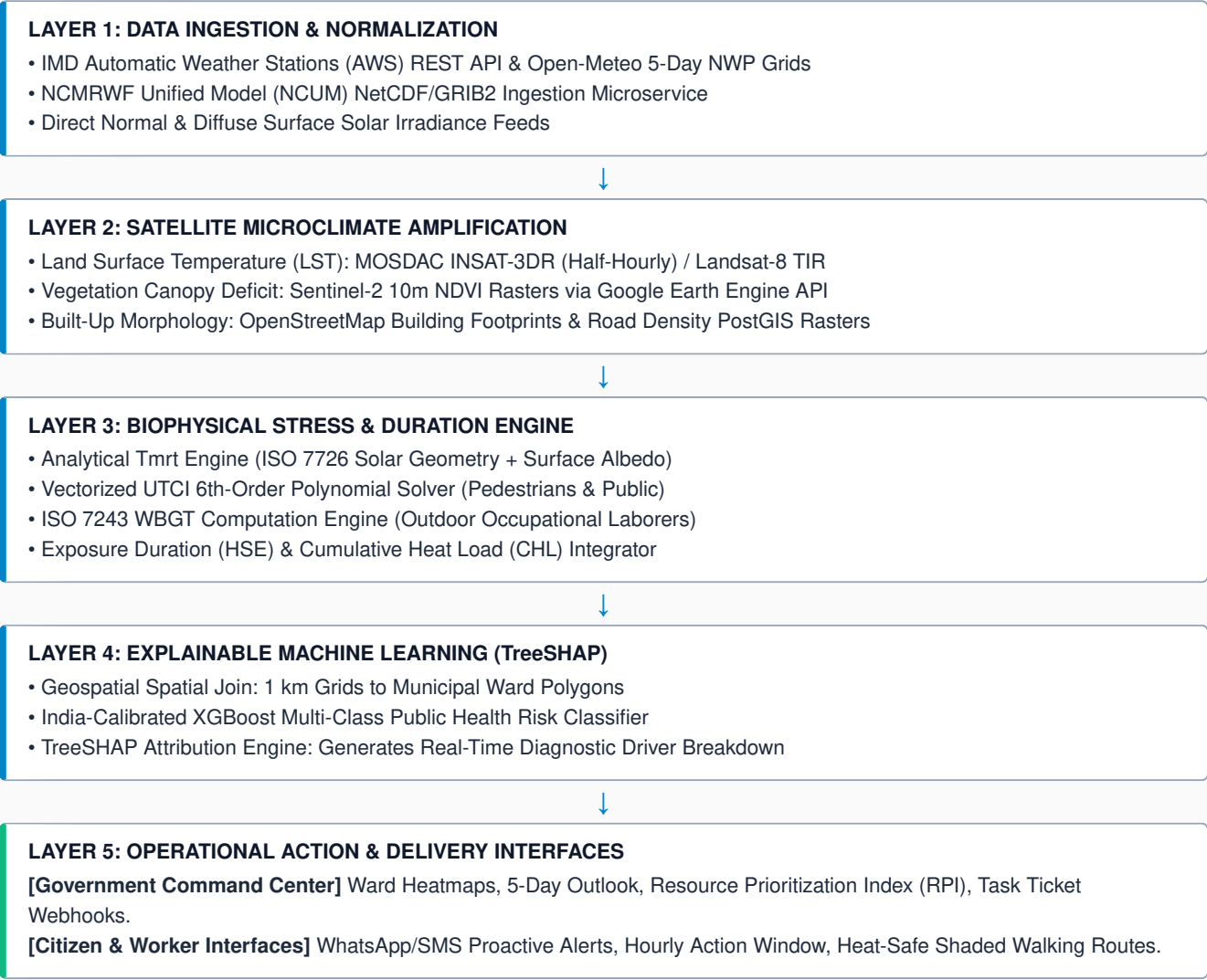
2.5 Cumulative Heat Load & Exposure Duration

Citing *Shah et al. (Nature Communications, 2025)*, prolonged heat exposure accumulates physiological load:

1. **Heat-Stress Exposure Duration (HSE):** $HSE_d = \sum I(UTCI_t \geq 38.0^{\circ}C)$ (Hours/day)
2. **Cumulative Heat Load (CHL):** $CHL_d = \sum \max(UTCI_t - 38.0, 0)$ (°C · hours)
3. **Warm Night Penalty (WNP):** Multiplier of 1.25x if minimum night temperature $T_{min} \geq 28.0^{\circ}C$ for ≥ 2 consecutive nights (prevents nocturnal cardiovascular recovery).

3. End-to-End System Architecture

SYSTEM ARCHITECTURE & DATA FLOW PIPELINE



4. The Machine Learning & Explainability Engine (TreeSHAP)

Feature Vector Formulation: For each ward w on forecast day t :

$$x_{w,t} = [\textit{UTCI}_{max}, \textit{WBGT}_{max}, \textit{LST}_{anomaly}, \textit{NDVI}_{mean}, \textit{BuildingDensity}, \textit{HSE}, \textit{CHL}, \textit{WNP}, \textit{SlumRatio}, \textit{ElderlyRatio}, \textit{LaborRatio}]$$

Model Architecture: XGBoost 2.0 multi-class classifier predicting 4 risk tiers (Low, Moderate, High, Extreme Emergency). Calibrated against 15 years of meteorological reanalysis and epidemiological heat-mortality baseline data.

TREESHAP FEATURE ATTRIBUTION BREAKDOWN (WARD 18 - OLD CITY)

Baseline Regional Heat Risk: +22%
+ High UTCI (44.6°C):  (+28%)
+ High Surface Temp (LST 51°C):  (+18%)
+ Severe Canopy Deficit (NDVI 0.08):  (+12%)
+ Building Concrete Density:  (+8%)
- Moderate Wind Speed (3.2 m/s):  (-4%)
+ Consecutive Warm Nights (>30°C):  (+4%)

PREDICTED RISK: 88% (EXTREME EMERGENCY)

5. Operational Municipal Dispatch & Action Engine

Resource Prioritization Index (RPI)

Ranks municipal wards objectively to allocate water tankers and medical teams during severe heatwaves:

$$RPI_w = [(UTCI_{max} - 26) / 20] \times PopDensity_w \times (0.40 \cdot SlumRatio + 0.35 \cdot ElderlyRatio + 0.25 \cdot LaborRatio) \times WNP \times (1 + 0.1 \cdot HSE_w)$$

Automated Municipal Task Ticket Payload (JSON)

```
{
  "ticket_id": "HAP-2026-083-W18-D2",
  "ward_id": "WARD_18",
  "ward_name": "Old City Bazaar",
  "priority_rank": 1,
  "risk_category": "EXTREME",
  "predicted_utci_peak": 44.6,
  "action_window": "11:30 - 16:00",
  "directives": [
    {
      "department": "Water_Supply_Sanitation",
      "action": "DEPLOY_MISTING_TANKERS",
      "quantity": 2,
      "target_location": "Old Market Chowk (23.0225° N, 72.5714° E)"
    },
    {
      "department": "Labor_Enforcement",
      "action": "MANDATORY_WORK_STOPPAGE",
      "target_group": "Outdoor Construction & Road Works",
      "effective_hours": "11:30 - 15:30"
    }
  ]
}
```

6. Citizen & Worker Interface (SMS/WhatsApp First)

WhatsApp / SMS Proactive Warning Broadcast



HEATSHIELD ALERT — WARD 18 (OLD CITY)

Condition: EXTREME HEAT STRESS Tomorrow (12:00 PM – 4:00 PM)

Physiological Risk: High danger of Heat Stroke & Dehydration for outdoor work.

CRITICAL SURVIVAL ACTIONS:

- 1. Shift heavy physical labor to morning hours (06:00 AM – 10:00 AM).
 - 2. Free air-cooled refuge open at: Municipal Hall (Old Market Gate 3).
 - 3. Free cold drinking water point: 150m away (Bazaar Square).
- Live Shaded Walking Route: <https://heatshield.in/w/18>

Heat-Safe Shaded Walking Route Recommender: Ranks pedestrian street network edges via cost function balancing distance against tree canopy coverage (NDVI) and surface temperature (LST):

$$EdgeCost = Distance \times [1 + 2.5 \cdot (1 - NDVI_{canopy})] \times (LST_{street} / 40)$$

7. Database Schema & API Specifications

```
-- Ward Forecast Master Table
CREATE TABLE ward_forecasts (
  forecast_id SERIAL PRIMARY KEY,
  ward_id VARCHAR(32) REFERENCES wards(ward_id),
  forecast_date DATE NOT NULL,
  lead_day INT NOT NULL CHECK (lead_day BETWEEN 1 AND 5),
  t_air_max FLOAT NOT NULL,
  utci_max FLOAT NOT NULL,
  wbgt_max FLOAT NOT NULL,
  hse_hours FLOAT NOT NULL,
  chl_load FLOAT NOT NULL,
  warm_night BOOLEAN DEFAULT FALSE,
  risk_category VARCHAR(16) NOT NULL,
  rpi_score FLOAT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Dispatch Audit Log Table
CREATE TABLE dispatch_tickets (
  ticket_id VARCHAR(64) PRIMARY KEY,
  ward_id VARCHAR(32) REFERENCES wards(ward_id),
  issued_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  action_type VARCHAR(64) NOT NULL,
  target_officer_phone VARCHAR(16) NOT NULL,
  dispatch_payload JSONB NOT NULL,
  delivery_status VARCHAR(32) DEFAULT 'SENT'
);
```

8. 36-Hour Hackathon Hour-by-Hour Execution Sprint

Timeline	Milestones & Team Deliverables (6-Member Division of Labor)
Hours 00 – 06 (Data & Contracts)	• Git repo initialized; PostGIS DB container spun up via Docker. • Ingest pilot city GeoJSON ward boundaries (Ahmedabad 48 wards). • Freeze JSON data contracts; implement Python UTCI & WBGT vectorized routines.
Hours 06 – 16 (Computation Core)	• Ingest Open-Meteo 5-day NWP forecasts (Temp, RH, Wind, Radiation). • Spatial join weather grids to ward polygons using GeoPandas. • Implement ISO 7726 analytical Tmrt engine; train baseline XGBoost classifier.

Timeline	Milestones & Team Deliverables (6-Member Division of Labor)
Hours 16 – 24 (UI & Mentoring)	• Build interactive MapLibre GL UI with choropleth ward risk layers. • Implement Ward Drill-Down Modal with dynamic TreeSHAP waterfall card. • Deploy RPI ranking table; present working math to Mentors in R1/R2.
Hours 24 – 32 (Action & Polish)	• Build Dispatch Webhook Engine (Twilio WhatsApp API / simulator). • Implement Citizen View (Hourly Action Window & Shaded Walking Route). • Seed simulation scenario: heat spike on Day +2 triggers auto-ticket.
Hours 32 – 36 (Pitch Rehearsal)	• Rehearse 5-minute live pitch and 60-second "Hero Demo" 10 times. • Prepare offline fallback video & local mock server in case of Wi-Fi drop.

9. The Winning 5-Minute Live Pitch & 60-Second "Hero Demo"

5-MINUTE LIVE STAGE PITCH TIMELINE

[0:00 - 0:45] THE HOOK (The Fatal Fallacy)
 Highlight why humidity makes 39°C in Kolkata more lethal than 49°C dry heat in Delhi (\$E o 0\$). Introduce HeatShield AI's physiological ward-level forecasting shift.

[0:45 - 1:30] THE SCIENCE & MATH
 Explain NCMRWF ingestion, ISO 7726 analytical Tmrt derivation, and dual UTCI/WBGT biophysical stress engines.

[1:30 - 2:45] THE 60-SECOND HERO DEMO
 1. Open Map & pan 48 wards → 2. Advance slider to Day +2 (Ward 18 flashes RED) → 3. Click Ward 18 for TreeSHAP diagnostic breakdown → 4. Enter judge's phone number and hit "Execute Emergency HAP Directive" → 5. Judge's phone buzzes on stage with live WhatsApp task ticket!

[2:45 - 4:00] SCALABILITY & MISSION MAUSAM FIT
 Highlight low operational cost (<₹4,500/month/city) and alignment with NDMA guidelines & Mission Mausam 2025–2026.

[4:00 - 5:00] Q&A DEFENSE
 Confidently transition to jury cross-questioning.

10. The "Anti-BS" Judge Defense Sheet (30 Technical Q&As)

Q1: How is this different from IMD's Heatwave Alerts?
 IMD alerts operate at district level using ambient shade temperature. We operate at 500m ward level, compute true physiological stress (UTCI/WBGT including humidity/radiation), and automate municipal dispatch.

Q2: How do you calculate Mean Radiant Temperature (\$T_{\{mrt\}}\$) without physical sensors?
 We implement the ISO 7726 analytical solar radiation balance equation, coupling satellite surface solar irradiance with sun-zenith geometry, ambient temperature, and surface albedo derived from Sentinel-2.

Q3: Why use UTCI instead of the NOAA Heat Index?
 The NOAA Heat Index was calibrated for indoor/shaded environments and errors out above 42°C in high humidity. UTCI is based on a full thermodynamic multi-node human heat budget valid across all extremes.

Q4: Why do you compute WBGT in addition to UTCI?

UTCI models pedestrian thermal comfort for the general public. WBGT is the global ISO 7243 standard for occupational labor, enabling legal enforcement of work-to-rest cycles for outdoor laborers.

Q5: Where do you get real-time mortality data to train your ML model?

Real-time death records take weeks to register digitally in India. We ground our risk categories in peer-reviewed Distributed Lag Non-linear Models (DLNM) from Ahmedabad/Delhi public health studies and backtest against historical mortality curves.

Q6: What machine learning algorithm do you use and why?

We use XGBoost 2.0 with TreeSHAP. Tree ensembles excel on tabular geospatial data, avoid overfitting, and natively compute exact Shapley values for explainability.

Q7: How do you reach illiterate outdoor laborers without smartphones?

We trigger automated voice-IVR calls in local languages and plain-text SMS via CDAC Mobile Seva, pushing alerts to registered labor union heads and ward corporators.

Q8: How does your platform fit into Mission Mausam (2025–2026)?

Mission Mausam enhances weather forecasting accuracy and decision impact. HeatShield AI fulfills the 'last-mile' requirement by translating NCMRWF model outputs into ward-level survival actions.

11. Team Member Stage Roles & Defense Responsibilities

Team Member Role	Live Defense Responsibility & Topic Mastery
Member 1: Team Lead <i>(Pitch Strategist)</i>	High-level vision, Mission Mausam fit, NDMA policy alignment. Controls live demo flow and conducts the phone-buzz hero moment on stage.
Member 2: Domain Scientist <i>(Biophysics Specialist)</i>	ISO 7726 Tmrt math, Stefan-Boltzmann energy balance equations, vapor pressure. Defends UTCI vs NOAA HI; explains clinical heat stroke dynamics.
Member 3: Geospatial Eng. <i>(GIS & Data Pipeline)</i>	NetCDF/GRIB2 ingestion, PostGIS spatial queries, R-Tree indexing. Sentinel-2 NDVI processing and INSAT-3DR LST spatial joins.
Member 4: ML Specialist <i>(ML & XAI Lead)</i>	XGBoost hyperparameter tuning, spatial cross-validation. TreeSHAP mathematical attribution and feature importance defense.
Member 5: Backend & Ops <i>(Systems Architect)</i>	FastAPI microservices, Celery async task queue, Redis caching. Webhook delivery mechanisms, Twilio/CDAC SMS gateway fallback.
Member 6: Frontend & UX <i>(UI/UX & Accessibility)</i>	MapLibre GL JS vector tile rendering, citizen PWA view. Hourly Action Window UI, shaded route recommender, offline UI.

HeatShield AI — SIH 2026 Definitive Technical Document

Mathematically Validated & Operationalized for 1st-Place Execution